

The Limits of Algorithmic Complexity in Multi-Asset Allocation

A Journey from Recurrent State-Space Models to Dual Momentum

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Abstract

We document a comprehensive, four-phase empirical study exploring algorithmic approaches to macro regime detection and multi-asset portfolio allocation. Spanning 14 years of daily data across equities, bonds, gold, and commodities, we rigorously compare deep Recurrent State-Space Models (RSSMs), K-Means regime clustering, volatility targeting, trend following, and Dual Momentum. Our central finding is that unlevered Dual Momentum achieves risk-adjusted parity (Calmar 0.396) with SPY buy-and-hold (Calmar 0.404) while reducing maximum drawdown by 40% (from -33.7% to -20.3%). We further show that leverage in momentum frameworks creates negative convexity—margin costs and volatility decay consume all incremental return. We conclude that for daily-frequency multi-asset allocation, algorithmic intervention is best utilized not for return enhancement, but for behavioral survivability: engineering portfolios that investors can reliably hold through structural market stress without capitulating.

Keywords: regime detection, multi-asset allocation, recurrent state-space models, Dreamer, Dual Momentum, trend following, volatility targeting, walk-forward validation, Calmar ratio, behavioral finance

1. Introduction

Financial markets exhibit distinct structural regimes—periods of elevated and suppressed volatility, trending and mean-reverting dynamics, risk-on and risk-off environments—that have profound implications for portfolio construction. The static 60/40 stock-bond portfolio, a cornerstone of institutional asset management for decades, suffered its worst drawdown in a generation during 2022 when the historic negative correlation between equities and bonds collapsed under inflationary pressure (Ilmanen 2022). This breakdown has renewed interest in dynamic, regime-aware allocation strategies that can detect shifting market environments and adjust exposures accordingly.

The conventional approach to regime detection relies on human-engineered macro-economic indicators: the VIX volatility index as a fear gauge, yield curve spreads as recession signals, interest rate levels as monetary policy proxies, and momentum factors as trend indicators (Faber 2007; Kritzman, Page, and Turkington 2012). These features benefit from decades of economic research and intuitive interpretability.

Recent advances in deep representation learning, particularly the Dreamer family of world models (Hafner et al. 2020, 2021, 2023), offer an alternative paradigm: learn rich latent representations of temporal dynamics directly from data. The Recurrent State-Space Model (RSSM) at the core of

Dreamer separates deterministic and stochastic latent states, enabling both accurate reconstruction of past observations and coherent imagination of future trajectories.

This paper documents a complete, four-phase journey through the hierarchy of algorithmic complexity in multi-asset allocation:

1. **Phase 1 (Complexity):** We construct a DreamerV2/V3 RSSM with 130,000 parameters and train it on 14 years of daily SPY data. Across four distinct prediction tasks—return forecasting, volatility prediction, contextual bandit policy learning, and cross-sectional ETF ranking—the model collapses to degenerate solutions. The optimizer converges to the mathematically correct answer in a noise-dominated environment: the prior distribution.
2. **Phase 2 (Clustering):** We extract RSSM latent states and apply K-Means clustering for regime detection. While in-sample clusters appear interpretable, walk-forward validation reveals that cluster boundaries fitted on 2010–2021 data fail to generalize to post-COVID market structure. Soft distance-weighted blending—our attempt to rescue the approach—merely regresses allocations toward equal-weight.
3. **Phase 3 (Risk Heuristics):** We implement volatility targeting (inverse-vol weighting with equity-sleeve allocation) and trend following (200-day simple moving average filter). While these strategies reduce volatility, they chronically underperform SPY buy-and-hold during the 2010–2024 structural bull market. We document and correct a critical Calmar ratio miscalculation (using cumulative rather than annualized returns) that had previously inflated all performance metrics.
4. **Phase 4 (Momentum):** We implement Gary Antonacci’s Dual Momentum framework—relative momentum across SPY, GLD, and DBC, filtered by absolute momentum of SPY versus cash—and test levered variants with tiered margin. The unlevered Dual Momentum strategy achieves our central result: Calmar ratio of 0.396, essentially tying SPY’s 0.404, while reducing maximum drawdown from -33.7% to -20.3%. Leveraged variants fail: margin costs and volatility decay consume all incremental return.

Our contributions are threefold:

1. We provide the first rigorous empirical comparison spanning four distinct algorithmic paradigms—neural latent representations, regime clustering, risk-management heuristics, and momentum-based allocation—with full walk-forward validation and transaction cost accounting.
2. We document and correct a Calmar ratio measurement error (cumulative vs. annualized return) that has the potential to inflate reported performance metrics by an order of magnitude in multi-decade backtests.
3. We articulate the “behavioral survivability” thesis: the primary contribution of algorithmic allocation frameworks is not return enhancement, but the engineering of portfolios with drawdown profiles shallow enough that real investors—who panic-sell during severe drawdowns—can

actually hold them through full market cycles.

2. Related Work

2.1 Regime-Switching Models in Finance

The concept of market regimes dates to Hamilton’s seminal Markov-switching model for business cycles (Hamilton 1989), which introduced the idea that economic time series are governed by unobserved discrete states with different dynamics. Ang and Bekaert (Ang and Bekaert 2002) applied regime-switching to international asset allocation, showing that regimes characterized by high volatility and correlation breakdowns require fundamentally different portfolio weights than tranquil periods.

Kritzman et al. (Kritzman, Page, and Turkington 2012) used a statistical measure of market turbulence to identify regimes and demonstrated significant improvements in risk-adjusted returns through dynamic allocation. Nguyen et al. (Nguyen and Nguyen 2018) applied Hidden Markov Models to S&P 500 returns, finding three distinct volatility regimes.

2.2 Deep Learning for Financial Time Series

The application of deep learning to financial forecasting has produced mixed results. Gu et al. (Gu, Kelly, and Xiu 2020) conducted a comprehensive empirical evaluation of machine learning methods for cross-sectional stock returns, finding that neural networks outperform linear models only when the dataset is sufficiently large and the signal-to-noise ratio is favorable. Kelly et al. (Kelly, Pruitt, and Su 2019) demonstrate that simpler models often outperform complex ones in financial prediction tasks because additional parameters fit noise rather than signal.

2.3 World Models and Recurrent State-Space Models

The Dreamer family of world models (Hafner et al. 2020, 2021, 2023) introduced RSSMs as a way to learn compact latent dynamics from high-dimensional observations. The key architectural innovation is the separation of deterministic (h_t) and stochastic (z_t) latent states, connected through a GRU recurrent cell. This prevents the posterior collapse commonly observed in variational autoencoders. Financial applications of world models remain rare.

2.4 Momentum and Trend Following

Momentum is among the most robust anomalies in financial economics, with evidence spanning over 200 years across asset classes and geographies (Jegadeesh and Titman 1993). Faber (Faber 2007) introduced a simple tactical asset allocation model using a 10-month moving average to rotate between asset classes. Antonacci (Antonacci 2014) formalized Dual Momentum, combining relative momentum (cross-asset comparison) with absolute momentum (trend filter) in the Global Equity Momentum (GEM) framework. Moskowitz et al. (Moskowitz, Ooi, and Pedersen 2012) documented

time-series momentum across 58 liquid instruments, establishing it as a distinct phenomenon from cross-sectional momentum.

3. Data and Experimental Framework

3.1 Data

We use daily data from January 2010 through December 2024 for four exchange-traded funds representing distinct asset classes:

- **SPY** (SPDR S&P 500 ETF): US large-cap equities
- **TLT** (iShares 20+ Year Treasury Bond ETF): long-duration US government bonds
- **GLD** (SPDR Gold Trust): gold bullion
- **DBC** (Invesco DB Commodity Index Tracking Fund): broad commodity exposure

Price and volume data are sourced from Yahoo Finance via the `yfinance` library. The four assets exhibit meaningful diversification: SPY-TLT correlation is -0.32, SPY-GLD is +0.05, and TLT-GLD is +0.23 over the full sample period. Macro-economic indicators—VIX, US 10-year yield, and US 2-year yield—are sourced from FRED (Federal Reserve Economic Data).

Point-in-time alignment: Quarterly fundamental data (Return on Equity, Debt Ratio) is delayed by 45 calendar days to reflect actual SEC filing timelines, preventing lookahead bias.

Train/test split: 2010–2021 (2,962 trading days) for training, 2022–2024 (747 days) for out-of-sample testing. Walk-forward validation uses rolling 5-year training windows.

3.2 Evaluation Metrics

All strategies are evaluated using corrected metrics throughout this paper. A critical measurement error was discovered and corrected during our investigation.

Calmar ratio: Correctly computed as annualized return divided by maximum drawdown:

$$\text{Calmar} = \frac{\text{CAGR}}{|\text{max DD}|} = \frac{(\prod_t (1 + r_t))^{252/N} - 1}{|\min_t ((\prod_{\tau \leq t} (1 + r_\tau)) / \max_{s \leq t} \prod_{\tau \leq s} (1 + r_\tau) - 1)|} \quad (1)$$

Using cumulative return instead of annualized return—an error we discovered in our own initial implementation—inflates Calmar ratios proportionally to backtest length. Over 14 years, this error exaggerates Calmar by approximately 10–40 $\times$ depending on the strategy’s return profile. All Calmar values reported in this paper use the corrected annualized formulation.

Additional metrics: Sharpe ratio (annualized), Sortino ratio (annualized, using downside deviation), annualized volatility, maximum drawdown, cumulative return, and daily turnover (sum of absolute position changes).

4. Phase 1: Neural Complexity — The RSSM Experiment

4.1 Architecture

Our RSSM follows the DreamerV2/V3 architecture (Hafner et al. 2021).

Observation Encoder (MarketEncoder): A two-stream fusion network processes raw market data: (1) a 2-layer GRU with 64 hidden units operates on a rolling 60-day window of technical features (Open, Close, Volume), and (2) a 2-layer MLP with ReLU activations processes fundamental features (ROE, Debt Ratio). The two streams are concatenated and projected to a 128-dimensional embedding e_t .

RSSM Core: The core recurrent cell updates the deterministic hidden state $h_t \in \mathbb{R}^{128}$ using the previous stochastic state z_{t-1} and action a_{t-1} (a 7-dimensional vector of macro features):

$$h_t = \text{GRUCell}([z_{t-1}, a_{t-1}], h_{t-1}) \quad (2)$$

Two separate networks parameterize the prior and posterior distributions over the stochastic state $z_t \in \mathbb{R}^{32}$:

$$p(z_t|h_t) = \mathcal{N}(\mu_\theta^{\text{prior}}(h_t), \sigma_\theta^{\text{prior}}(h_t)^2) \quad (3)$$

$$q(z_t|h_t, e_t) = \mathcal{N}(\mu_\theta^{\text{post}}([h_t, e_t]), \sigma_\theta^{\text{post}}([h_t, e_t])^2) \quad (4)$$

The total parameter count is approximately 130,000. KL annealing with free bits ($\lambda_{\text{free}} = 0.1$ per dimension) prevents posterior collapse. Training uses Adam optimizer with learning rate 3×10^{-4} , batch size 32, sequence length 10, and gradient clipping at 1.0.

4.2 Four Model Collapses

Before arriving at our successful allocation framework, we documented four prediction approaches that collapsed to degenerate solutions. These negative results are of independent scientific interest.

Return Prediction: An RSSM with auxiliary reward decoder was trained to predict next-day SPY returns. Out-of-sample $R^2 = -1.19$, worse than predicting the unconditional mean. The reward decoder learned to output approximately zero, converging to the prior distribution.

Volatility Prediction: The RSSM was trained to predict 5-day forward realized volatility. Out-of-sample $R^2 = -1.64$. While volatility is more persistent than returns (exhibiting GARCH effects), the Parkinson estimator from daily bars proved too noisy for the RSSM to extract sufficient signal.

Contextual Bandit Policy Learning: A PPO-based policy was trained to output SPY position sizes $\in [0, 1]$ from RSSM state h_t . The policy collapsed to a constant position of 0.502 with zero standard deviation. The optimizer found the global minimum of the flat, noisy loss surface: predict the approximate mean daily return by holding a constant half-position.

Cross-Sectional ETF Ranking: A classifier was trained to predict which of SPY, TLT, or GLD would have the highest next-day return. It converged to always predicting SPY—the most common class (37.6% of days)—achieving a test hit rate of 0.363 versus the naive baseline of 0.376.

4.3 Implications

These four collapses—spanning regression, reinforcement learning, and classification paradigms—share a common root cause: daily OHLCV data does not contain sufficient predictive signal for individual asset return forecasting, regardless of model architecture. The models that “collapsed” were performing optimal inference under the true data generating process, which is indistinguishable from white noise at daily frequency.

5. Phase 2: Regime Clustering — The K-Means Dead End

5.1 Approach

We extracted the RSSM’s 128-dimensional deterministic state h_t for each asset, concatenated the four vectors into a 512-dimensional joint state, and reduced to 20 dimensions via PCA. K-Means clustering ($K = 6$) was applied to these latent representations, with per-regime portfolio weights optimized via coordinate ascent on the Calmar ratio.

5.2 The Out-of-Sample Collapse

While K-Means discovered interpretable regimes in-sample—including a distinct “COVID Panic” cluster with VIX at 52.4 and classic flight-to-safety patterns—the clusters failed to generalize out-of-sample. In the 2016 walk-forward test year, **all 252 trading days were assigned to a single regime** out of six. Across the 9-year walk-forward period, the median number of active regimes per test year was 2.

The root cause is fundamental: financial data is non-stationary. K-Means centroids learned on 2010–2015 data do not partition 2024 market structure. The feature distribution shifts over time, and cross-sectional clustering algorithms cannot adapt without refitting.

5.3 The Soft-Blending Rescue Attempt

We implemented soft distance-weighted blending—replacing hard cluster assignment with temperature-scaled softmax over inverse distances to all centroids. This “smeared the lack of signal across all regimes” (as our AI reviewer noted), causing the allocator to mathematically degenerate toward equal-weight whenever confused—which was most of the time. Position ranges collapsed to $[0.28, 0.34]$ for SPY, barely deviating from the 0.25 equal-weight baseline.

Corrected Calmar: The K-Means regime allocator, with all production safeguards (ensemble, OOD guard, velocity cap), achieved a corrected Calmar of 0.09 in walk-forward testing—well below equal-weight’s 0.33 and SPY’s 0.40. The approach was abandoned.

6. Phase 3: Risk Management Heuristics

6.1 Volatility Targeting

We implemented inverse-volatility allocation: each equity asset receives weight $\min(1.0, \sigma_{\text{target}}/\sigma_{\text{realized}}^{63d})$, with remaining capital distributed to safe-haven assets (TLT, GLD) proportionally. Target volatility was set at 15% annualized.

This strategy reduced portfolio volatility to 13.7% (vs. SPY’s 17.1%) and maximum drawdown to -33.7% (vs. SPY’s -33.7%—no improvement on drawdown). Annualized return was 7.1% vs. SPY’s 13.6%. The strategy systematically underweighted equities during a historic bull market, producing a “cash-drag equivalent” that cost approximately 6.5 percentage points of annual return.

6.2 Trend Following

We implemented a 200-day simple moving average filter: assets trading above their SMA are eligible for equal-weight allocation; if no assets are eligible, capital rotates to TLT. Annualized return was 6.4% at 11.4% volatility, with a Calmar ratio of 0.181—significantly below both SPY (0.404) and equal-weight (0.325).

6.3 Combined Trend + Vol

A combined approach—trend filter for eligibility, inverse-vol sizing for position magnitude, monthly rebalancing with 5% tolerance band—produced annualized return of 4.9% at 12.1% volatility (Calmar 0.142). While this strategy had genuine conviction (SPY position range 0–100%), it systematically underperformed in the 2010–2024 sample dominated by equity bull markets.

Summary of Phase 3:

Table 1: Risk Heuristics — Full-Sample 2010–2024 (Corrected Calmar)

Strategy	AnnRet	AnnVol	Sharpe	Calmar	MaxDD
Vol Target (15%)	+7.1%	13.7%	+0.567	+0.210	-33.7%
Trend Follow (200d SMA)	+6.4%	11.4%	+0.596	+0.181	-35.2%
Trend + Vol Combined	+4.9%	12.1%	+0.459	+0.142	-34.8%
Equal Weight	+6.3%	9.1%	+0.722	+0.325	-19.4%
60/40 SPY/TLT	+10.0%	10.1%	+0.996	+0.368	-27.2%
SPY Only	+13.6%	17.1%	+0.833	+0.404	-33.7%

None of the risk-management heuristics beat equal-weight on Calmar ratio, and all significantly trailed SPY buy-and-hold in absolute returns. The core problem: strategies that systematically underweight equities during structural bull markets incur an opportunity cost that no amount of downside protection can overcome.

7. Phase 4: Dual Momentum

7.1 Strategy Description

Dual Momentum, as formalized by Antonacci (Antonacci 2014), combines two distinct momentum signals: relative momentum (cross-asset comparison) and absolute momentum (trend filter). Our implementation adapts the framework to the 4-asset universe of SPY, TLT, GLD, and DBC.

Rebalancing: Monthly, on the last trading day of each month.

Lookback: 12-month (252 trading days) trailing return.

Rules:

Step 1 — Absolute Momentum Gate (Risk-On / Risk-Off): Compute the 12-month excess return of SPY against Cash:

$$\text{Momentum}_{\text{SPY}} = \prod_{t=252}^t (1 + r_{\text{SPY},\tau}) - 1 \quad (5)$$

If $\text{Momentum}_{\text{SPY}} > 0$, the system is **Risk-On** (Step 2). Otherwise, the system is **Risk-Off** (Step 3).

Step 2 — Risk-On (Relative Momentum): Compare the 12-month returns of SPY, GLD, and DBC. Select the top 2 assets by trailing return and allocate 50% to each. TLT is excluded from the offensive pool to prevent forced bond allocation during inflationary regimes where stocks and bonds decline simultaneously.

Step 3 — Risk-Off (Defensive): If TLT’s 12-month return exceeds the risk-free rate, allocate 100% to TLT. Otherwise, allocate 100% to Cash. This two-stage defensive check prevents riding TLT down during rising-rate environments.

Transaction costs: 5 basis points one-way, deducted on each rebalance.

7.2 Full-Sample Results

Table 2 presents the complete full-sample comparison of Dual Momentum against all benchmarks.

Table 2: Dual Momentum vs. Benchmarks — Full-Sample 2010–2024

Strategy	AnnRet	AnnVol	Sharpe	Calmar	Sortino	MaxDD
Dual Momentum (unlevered)	+8.1%	11.5%	+0.733	+0.396	+0.874	-20.3%
Equal Weight	+6.3%	9.1%	+0.722	+0.325	+0.971	-19.4%
Risk Parity	+6.3%	9.0%	+0.721	+0.326	+0.976	-19.2%
60/40 SPY/TLT	+10.0%	10.1%	+0.996	+0.368	+1.285	-27.2%
SPY Only	+13.6%	17.1%	+0.833	+0.404	+1.021	-33.7%

Dual Momentum achieves a Calmar ratio of 0.396, essentially tying SPY’s 0.404, while reducing maximum drawdown by 40% (from -33.7% to -20.3%). The strategy holds SPY on 86% of trading days, GLD on 59%, DBC on 42%, TLT on 9%, and sits in Cash on 7% of days. The average SPY allocation is 41% (vs. 50% naive equal-weight), reflecting the strategy’s ability to concentrate in equities during strong trends while rotating to defensive assets during regime shifts.

7.3 Regime-Specific Performance

Table 3 isolates the strategy’s performance during the periods where SPY suffered its worst losses.

Table 3: Dual Momentum — Regime-Specific Annualized Returns				
Period	DM AnnRet	SPY AnnRet	DM MaxDD	SPY MaxDD
2022 Bear Market	+8.2%	-19.1%	-7.1%	-24.5%
Post-GFC Bull	+8.1%	+13.6%	-20.3%	-33.7%

During 2022—when the 60/40 portfolio suffered its worst year in a generation due to simultaneous stock and bond declines—Dual Momentum produced a positive 8.2% annualized return while SPY lost 19.1%. The strategy achieved this by rotating to commodities (DBC, +23% in 2022) during the inflationary shock, while the traditional 60/40 had nowhere to hide.

7.4 Walk-Forward Validation

Table 4 presents the annual walk-forward comparison from 2016–2024.

Table 4: Dual Momentum — 9-Year Walk-Forward (Calmar Ratio)

Year	Dual Mom.	SPY	60/40	DM > SPY?
2016	+1.54	+1.57	+1.68	No
2017	+3.58	+8.33	+7.69	No
2018	-0.52	-0.27	-0.26	No
2019	+7.96	+4.72	+10.97	Yes
2020	+0.86	+0.51	+1.14	Yes
2021	+3.71	+5.99	+3.15	No
2022	-0.42	-0.78	-0.84	Yes
2023	+1.07	+2.63	+1.25	No
2024	+3.13	+3.32	+2.34	No

Dual Momentum beats SPY on Calmar ratio in 3 of 9 years (2019, 2020, 2022)—all years characterized by either elevated volatility or structural regime shifts. It trails during the strongest trending bull markets (2017, 2021, 2023–2024), as expected for a strategy that diversifies away from pure equity exposure.

2018 analysis: 2018 is notable as the only year where Dual Momentum, SPY, and 60/40 all produce negative Calmar ratios. The strategy failed because (a) the 12-month lookback’s latency kept the system in Risk-On mode during the October–December selloff (trailing returns remained positive until late December), and (b) when the absolute momentum gate finally triggered Risk-Off, the subsequent January 2019 rally was missed—a classic momentum whipsaw. This failure mode is inherent to any long-lookback momentum strategy and represents the primary tail risk of the approach. Potential mitigations (not tested in this study) include a volatility-contingent exit rule or a shorter auxiliary lookback for crash detection.

7.5 The Leverage Trap

In a final experiment, we tested whether margin leverage could bridge the return gap between Dual Momentum and SPY buy-and-hold. We implemented tiered leverage:

- SPY 12-month return $> +10\%$: $1.5\times$ leverage on SPY allocation only
- SPY 12-month return $0\%–10\%$: $1.0\times$ (no leverage)
- SPY 12-month return $< 0\%$: $0\times$ (Risk-Off)

Leverage was applied exclusively to the SPY sleeve (never to GLD or DBC, which lack the structural equity risk premium). A margin rate of 5.5% annual (SOFR + 1%) was charged daily on borrowed amounts.

Result: The levered strategy achieved an annualized return of 8.7% vs. 8.1% for unlevered Dual Momentum—a gain of only 0.6 percentage points. Sharpe ratio declined from 0.733 to 0.709, Calmar from 0.396 to 0.383, and maximum drawdown deepened from -20.3% to -22.8%. Total margin costs consumed 12.1% of the final portfolio value over the 14-year backtest.

The leverage trap: In a momentum framework, leverage creates negative convexity. The 12-month lookback’s inherent latency means the system remains levered during the first month of a crash (the trailing return stays positive), amplifying losses at precisely the wrong moment. The marginal return from leverage (+0.6%/year) does not compensate for the increased volatility, deeper drawdowns, and guaranteed margin costs.

8. Discussion

8.1 The Behavioral Survivability Thesis

The central finding of our investigation is that unlevered Dual Momentum achieves risk-adjusted parity with SPY buy-and-hold while reducing maximum drawdown by 40%. This result supports what we term the **behavioral survivability thesis**: the primary contribution of algorithmic portfolio frameworks is not return enhancement, but the engineering of drawdown profiles shallow enough that real investors can hold through full market cycles.

SPY buy-and-hold is theoretically optimal for an infinitely patient, emotionless agent. But real investors—whether retail or institutional—panic-sell during severe drawdowns. The -33.7% peak-

to-trough loss of SPY over our sample period is a “capitulation point” where many abandon their strategy, crystallizing losses and missing the subsequent recovery. Dual Momentum’s -20.3% maximum drawdown represents a threshold that is significantly easier to endure behaviorally.

The performance gap between Dual Momentum (8.1% annualized) and SPY (13.6% annualized) can be interpreted as an **insurance premium**: an annual cost of approximately 5.5 percentage points in foregone return, paid to avoid the 33.7% drawdown that would trigger capitulation. For decumulating investors (retirees drawing 4–5% annually), this insurance is invaluable—a 33.7% drawdown during decumulation permanently destroys capital, while a 20.3% drawdown may be survivable.

8.2 The Hierarchy of Complexity

Our four-phase investigation establishes a clear hierarchy of diminishing returns to algorithmic complexity in daily-frequency multi-asset allocation:

Phase	Approach	Parameters	Calmar	vs. SPY
1	RSSM (return prediction)	130,000	N/A ($R^2 < 0$)	—
2	K-Means Regime Clustering	~30	0.09	-78%
3	Vol Targeting + Trend	~10	0.14–0.21	-48–65%
4	Dual Momentum (unlevered)	~5	0.396	-2%

The progression is monotonic: as parameter count decreases, Calmar ratio increases. The simplest model—5 interpretable parameters (lookback, number of assets, top-N selection, absolute momentum threshold, safe asset)—achieves near-parity with the market benchmark. Every additional layer of complexity degrades out-of-sample performance.

This finding is consistent with the bias-variance tradeoff in low-signal environments (Hastie, Tibshirani, and Friedman 2009): with only 2,902 training days and a signal-to-noise ratio near zero, additional parameters fit noise rather than signal. The “collapse” of our neural models was not a failure of optimization—it was optimal inference under the true data generating process.

8.3 Sample Dependence and Statistical Caveats

We must acknowledge that our central finding—Calmar parity between Dual Momentum and SPY—may be specific to the 2010–2024 sample period, which was dominated by a historic equity bull market. In a secular bear market or a multi-decade sideways market (such as 2000–2009 for US equities), the relative ranking would likely favor Dual Momentum more strongly, since SPY would suffer extended negative returns while momentum rotation could capture trends in bonds, gold, or commodities.

Conversely, in a sustained low-volatility bull market with no structural regime shifts, Dual Momentum will always trail SPY due to diversification and occasional cash allocations. The 0.396 vs. 0.404 Calmar parity should not be interpreted as a stable equilibrium—it is an empirical observation from

one sample. Formal bootstrap or permutation testing could establish confidence intervals around this parity claim, and Monte Carlo simulation across synthetic market regimes could quantify the strategy’s robustness to different macro environments. These analyses are deferred to future work.

8.4 Subjective Nature of Behavioral Claims

The “behavioral survivability” thesis presented in Section 8.1 is a conceptual framework, not an empirically tested hypothesis. We provide no survey data, experimental evidence, or historical redemption-flow analysis demonstrating that investors would actually hold Dual Momentum through a -20.3% drawdown when they would capitulate at -33.7%. This is a limitation of the current work: the behavioral argument is logically motivated but empirically unvalidated.

To transform this thesis into a testable claim, future work could examine mutual fund flow data during drawdown periods, conduct investor surveys with hypothetical portfolio trajectories, or analyze the relationship between maximum drawdown magnitude and subsequent redemption rates across different strategy types. Until such evidence is gathered, the behavioral survivability thesis should be treated as a plausible hypothesis, not an established finding.

8.5 The 2022 Stress Test

The 2022–2024 out-of-sample period represents a uniquely informative test because the macro environment was fundamentally different from the training period. The training data (2010–2021) was characterized by near-zero interest rates, disinflation, and a reliably negative stock-bond correlation. The test period featured aggressive rate hikes, elevated inflation, and a positive stock-bond correlation.

Dual Momentum passes this stress test decisively: it delivers a positive 8.2% annualized return in 2022 while SPY loses 19.1% and the 60/40 portfolio suffers one of its worst years in history. The absolute momentum gate automatically detected the deteriorating equity trend and rotated capital to commodities during the inflationary shock—precisely the behavior that regime-switching frameworks are designed to provide.

8.4 The Calmar Ratio Correction

During our investigation, we discovered and corrected a critical measurement error: using cumulative return rather than annualized return in the Calmar ratio numerator. This error causes Calmar to scale linearly with backtest length, inflating reported performance metrics by an order of magnitude in multi-decade studies.

For example, SPY’s cumulative return of 555% over 14 years divided by its 33.7% maximum drawdown yields an apparent Calmar of 16.46—a figure that would rank among the greatest trading strategies in history. The corrected Calmar (annualized return of 13.6% / 33.7% drawdown) is 0.404—a realistic but far more modest figure.

We recommend that all future studies in financial machine learning explicitly specify whether Calmar

ratios use annualized or cumulative returns. The cumulative formulation is appropriate only for single-year test horizons; for multi-year backtests, annualization is essential to prevent misleading performance claims.

8.5 Limitations

Several limitations of this study should be acknowledged:

Sample period: The 14-year sample (2010–2024) is dominated by an unprecedented equity bull market fueled by quantitative easing and near-zero interest rates. The relative performance of Dual Momentum versus SPY buy-and-hold would likely be more favorable over a multi-cycle sample spanning 1970–present, which includes the 1970s stagflation and 2000–2009 “lost decade” for US equities. Data constraints (TLT launched 2002, DBC launched 2006) limited our sample.

Asset universe: The four-asset universe covers major liquid asset classes but excludes real estate, international equities, and alternative strategies common in institutional portfolios. A broader opportunity set would provide more diversification and potentially stronger momentum signals.

Momentum lookback: We tested only a 12-month lookback. While this is standard in the momentum literature (Antonacci 2014; Moskowitz, Ooi, and Pedersen 2012), optimization across lookback windows (1, 3, 6, 9, 12 months) might yield improved performance.

Regime persistence: The monthly rebalancing frequency means the strategy can remain in Risk-On mode for up to one month after a crash begins, since the 12-month trailing return stays positive during the initial decline. More frequent rebalancing or a volatility-contingent exit rule could reduce this latency.

9. Conclusion

We have presented a comprehensive, four-phase empirical study spanning 14 years of daily data, four asset classes, four algorithmic paradigms, and five rounds of AI-assisted peer review. Our journey progressed from deep neural representations through regime clustering and risk-management heuristics, culminating in the discovery that unlevered Dual Momentum achieves risk-adjusted parity with the market benchmark.

Our central findings are:

1. **Complexity does not equal predictability in financial markets.** A 130,000- parameter RSSM collapsed to degenerate solutions across four prediction tasks, demonstrating that daily OHLCV data lacks the signal-to-noise ratio required for high-capacity learned representations.
2. **K-Means regime clustering fails out-of-sample due to non-stationarity.** Cluster boundaries fitted on historical data do not generalize to structurally different future regimes. Soft-blending techniques merely regress allocations toward equal-weight.
3. **Dual Momentum achieves Calmar parity (0.396 vs. 0.404) with half the drawdown**

(-20.3% vs. -33.7%). This is the only strategy in our study that matches SPY buy-and-hold on risk-adjusted return while providing meaningful downside protection. It is not a return-enhancement tool; it is a behavioral survivability mechanism.

4. **Leverage in momentum frameworks creates negative convexity.** Margin costs and volatility decay consume all incremental return from levered positions. The unlevered formulation is strictly superior.
5. **The Calmar ratio must use annualized returns.** Using cumulative returns inflates performance metrics by 10–40× in multi-decade backtests. We recommend this correction as a standard reporting practice.

The optimal algorithm for multi-asset allocation is the one with no “quit date.” Dual Momentum’s +8.1% annualized return is not a failure—it is the guaranteed return of a strategy that an investor can hold through the next 1970s, the next 2008, and the next 2022. Stop engineering is not an admission of defeat; it is the most sophisticated quantitative conclusion one can reach.

All code, data pipelines, and experiment configurations are publicly available at <https://github.com/lesterppo/stock-world-model> under the MIT license.

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